Self-repairing body armor
Mussels grip rocks with self-healing stretchy fibres, which has inspired an extremely tough and elastic new material. **http://dgit.in/MBdyArmr**

Racoon Dinosaur
A "masked" dinosaur from 130 million years ago could hide even in broad daylight from its predators. **http://dgit.in/RcnDino**

cause the patient is actually partially awake in many ways, but they may not be aware. Of course it's us laymen who use the term vegetative state – medicine calls it unresponsive wakefulness syndrome.

Patients in a vegetative state are often able to open their eyes, swallow, and even track moving objects with their eyes. There is rarely any other movement. They may or may not have normal sleep-wake cycles, and often can even make sounds, and they rarely need life support. Sadly, once a patient is classified as a persistent vegetative state, recovery is very, very rare. Most people who become vegetative and then recover, do so within days or weeks. The longer a patient is in a vegetative state, the less the chances of recovery.

Patients can live in such a state for weeks, months, years and in some cases even decades! India, shamefully, holds the record for the longest lived patient in a vegetative state in the world. In 1973, Aruna Shanbaug was a 25-year old nurse working at KEM hospital in Mumbai (Bombay, back then). On November 27, 1973, a cleaner working for the hospital attacked her, raped her and strangled her with metal chains, and left her to die. She didn't die, and was found and admitted to the ICU. Because of the strangulation she suffered brain damage, and was left in a persistent vegetative state. She was force fed through her nose, twice a day, in order to keep her alive, and was looked after by the staff in the hospital, for 42 years! She finally breathed her last in May 2015 after a short battle with pneumonia.

Because of Indian laws, despite several attempts by people supporting euthanasia, the courts always disallowed it. Many NGOs and journalists attempted to get the court to allow the painful force feeding to be stopped, so she could slip away peacefully, but always lost the case in court. If that wasn't bad enough, because the law didn't consider consider putting someone in a vegetative state to be akin to murder, her attacker, Sohanlal Bharta Walmiki was only charged with attempted murder. The rape was not mentioned on the report, perhaps because the Dean of KEM wanted to protect her as she was scheduled to be married soon, but it also meant that the accused was not charged with rape or sexual assault. In a mere 7 years Walmiki was released from jail, and then disappeared. Only after her death in 2015 was he tracked down by journalists to a village in UP, where he was living with his family, and is reported to be deeply regretful of his actions. If nothing else, at least Aruna's tragedy sparked a debate that eventually ended up with the supreme court allowing passive euthanasia (stopping medical treatment, or life support).

Whether or not patients in vegetative states are aware of their state is not known, and there are reports that suggest that it varies from patient to patient, but it is widely thought to be unlikely.

LOCKED IN

This is the stuff that all of us will have nightmares about! Although rare, there are cases of humans who have suffered total loss of bodily control, but are still as conscious and aware as a normal person. Most locked in patients are able to move their eyes and blink, and they have to communicate in that manner (blink once for yes, twice for no, etc.). The most famous person we all know of in this state is the physicist Stephen Hawking. As horrible as it already is, to imagine our perfectly healthy brains living in a body that just doesn't work, even Hawking would consider himself lucky to not be in a "totally locked in syndrome", also known as CLIS (completely locked in state). A CLIS patient is totally aware, but is unable to even move their eyes or blink, and it's unclear if they're able to even see.

It's truly a horrible thought… imagine waking up to complete darkness, unable to move, speak or do anything except think, and then have to live like that for months, or years! Such patients can easily be diagnosed as coma patients (and many in the past probably were), because they seem to be totally unresponsive to everything. It's scary to think how many people would have been thought to be vegetative in the past, but might actually just have been locked-in or completely locked-in! Worse still if the patient is able to hear and feel but not communicate or move!

What it all boils down to is that we're pretty ignorant still about how the brain works, and even where our consciousness is, and we have a long, long way to go before we can do away with the nightmare scenarios presented in this article, and help the people who inevitably end up like this.

BRAIN-COMPUTER INTERFACES

This is one solution to finding out where we go when we sleep, or seeing if a coma patient is still in there, or even helping vegetative state patients, and more importantly locked-in or completely locked-in patients. If we will eventually be able to translate brain activity, we might be able to look into the minds of such patients. Better yet, if we are able to set that up to work as a two way radio, and inject electrical signals into such a patient's brain, we might be able to set up communication. We're a long, long way away from achieving this though… or are we? Check out the Future Feature in our Alt section if you're interested in knowing more…